

INVX DGO Onboarding Program – Metadata

10:00 – 11:00 | 3 Oct 2023



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THERE ARE 2 TYPES OF METADATA – BUSINESS AND TECHNICAL

Metadata is information about data (e.g., definition, name, format, storage location)



Business Metadata

Information that describes the **process, context and business meaning about the data** within Group Data Zone (GDZ) and Monoline Data Zone (MDZ)

Examples of business metadata:

- Business Attribute Name
- Business Definition
- Column Name*
- Data Owner
- Last Update details



Technical Metadata

Information that describes **technical terms of data, its mapping and transformations** from source systems to the DataX (both GDZ and MDZ)

Examples of technical metadata:

- Schema/ Database Name
- Data Set Name
- Data Zone (MDZ/GDZ)
- Last Update details
- Column Name*

* The column name will be used for linkage with technical metadata.

BENEFITS OF METADATA MANAGEMENT FOR DATAX AND SUBSIDIARIES



Enable data users in DataX to **easily retrieve data** for their day-to-day tasks



Allow **deeper value generation** from data sets within DataX



Facilitate communication between DataX and Subsidiaries



Help manage data to **comply with regulatory requirements** for DataX and Subsidiaries

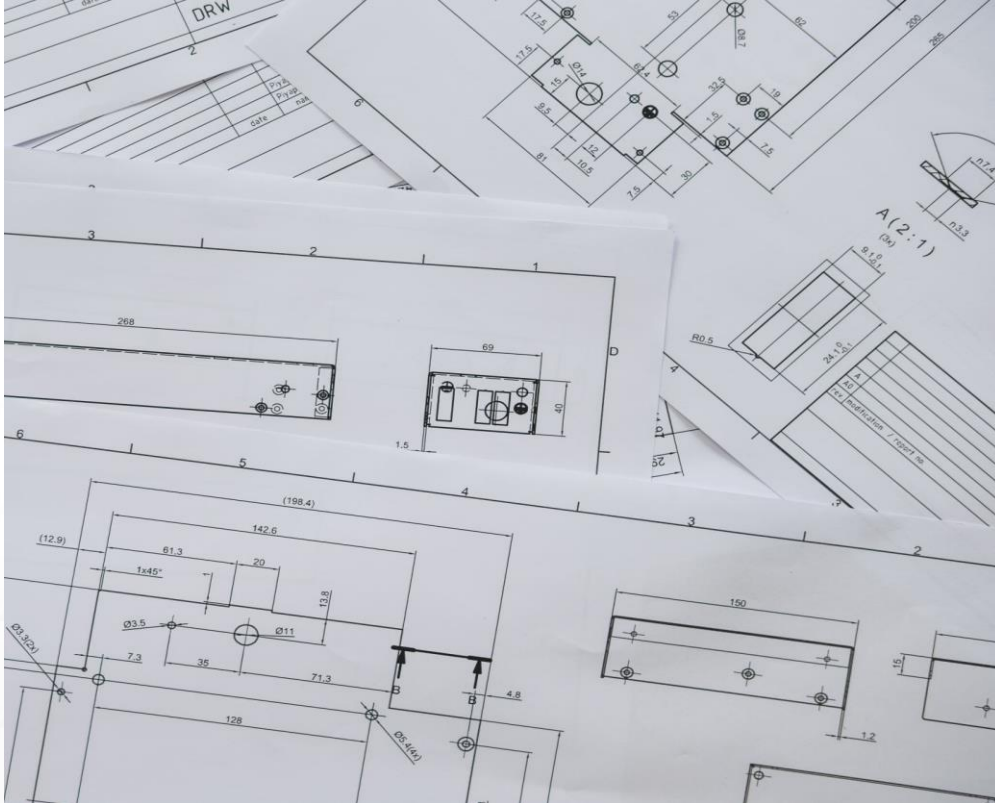


Ease of mapping, transferring and testing data for regular **maintenance** within DataX



Provide **accuracy and acceleration** in project delivery for data movement from SDZ to GDZ

THE METADATA STANDARD IS ESTABLISHED TO ENSURE CONSISTENT NAMING CONVENTIONS ARE APPLIED TO METADATA



Metadata Naming Convention for Various Fields within Business and Technical Metadata

- The naming convention for each field within the business and technical metadata template will be **concise, clear, accurate, and self-explanatory**.
- There will be a **unique business attribute name and business definition** in business metadata.
- Business and Technical Metadata will be **adequately detailed** to understand the data and improve accessibility of metadata by all users across DataX and subsidiary.
- Business and Technical Metadata should be **complete, accurate, concise, and up to date**.

RULES FOR METADATA MANAGEMENT

Business Metadata

- Business attribute name and business definition must have **1:1** mapping of each subsidiary
 - Column name will be **unique ID** for business metadata of each subsidiary
 - The same business attribute name **cannot occur** in multiple rows within each subsidiary
- Column name will be used for **linkage** with technical metadata

Technical Metadata

- Data subject area is included for **curated zone only**
- **Unique ID** for technical metadata will be a combination of source system fields such as **Schema/Database Name + Data Set Name + Column Name**
- Column name and system/ data subject area may have **many:1** mapping (Each system/ data subject area can contain more than 1 columns)
- The same name of column can be contained in more than 1 data set and data subject area.
- Column name **can occur** multiple times (multiple rows) in technical metadata. This implies that business metadata and technical metadata will be **1:many** for a column name
- In operationalization phase, metadata tool will be leveraged for automation so that technical metadata can be populated, with limited human intervention

METADATA MANAGEMENT – ROLES & RESPONSIBILITIES

DataX

Group Data Zone Data Steward

- DataX business data steward acts on behalf of DataX data owner and supports business metadata for respective data in group data zone (GDZ)
- DataX technical data steward supports technical metadata maintenance for respective data in group data zone (GDZ)

DataX Metadata Management Team

- DataX Metadata management team defines Metadata standards and processes for group data zone (GDZ) and subsidiary data zone (SDZ)
- DataX Metadata management team takes responsibility to review, register and/or update metadata for group data zone (GDZ)

Subsidiary

Subsidiary Data Steward

- Subsidiary business data steward acts on behalf of subsidiary data owner and supports business metadata for respective data in group data zone (GDZ) and subsidiary data zone (SDZ)
- Subsidiary technical data steward supports technical metadata maintenance for respective data in group data zone (GDZ) and subsidiary data zone (SDZ)

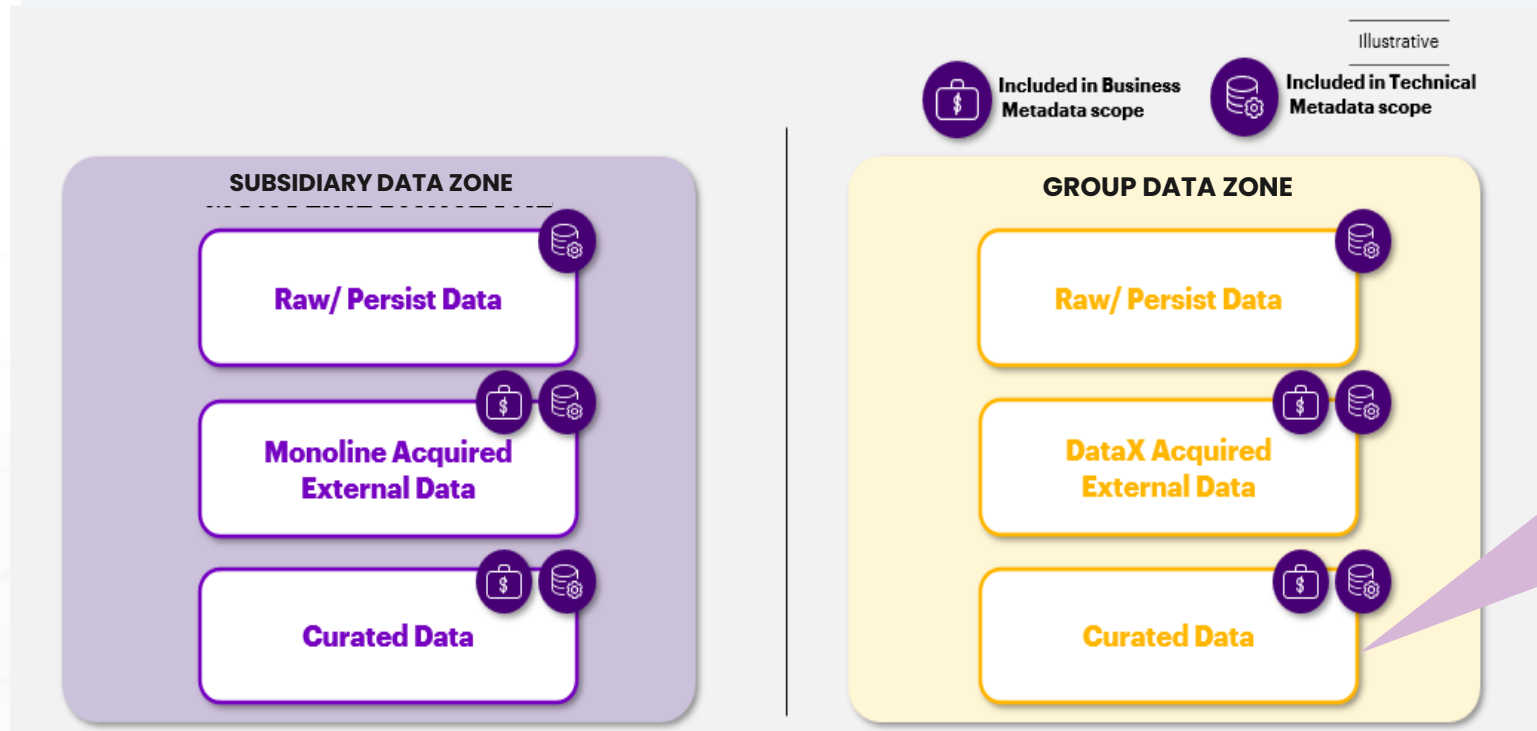
Subsidiary Metadata Management Team

- Subsidiary Metadata management team takes responsibility to review, register and/or update metadata for data in group data zone (GDZ) and subsidiary data zone (SDZ)

Data Gov. Components	Roles and Responsibilities	Group Data Zone		Subsidiary Data Zone	
		DataX	Subsidiaries	DataX	Subsidiaries
Metadata Management	• Define Business & Technical Metadata Management Documentation Processes & Standards	✓		✓	
	• Document & maintain Business Metadata following defined processes	✓ (external, aggregated data)	✓ (subsidiary data)		✓
	• Document & maintain Technical Metadata following defined processes	✓ (external, aggregated data)	✓ (subsidiary data)		✓

SCOPE OF BUSINESS METADATA DEVELOPMENT IN THE GROUP DATA ZONE (GDZ) AND SUBSIDIARY DATA ZONE (SDZ)

In the context of DataX, the scope of business metadata includes metadata for all data sets in the Subsidiary Data Zone (SDZ) and the Group Data Zone (GDZ) except for persist data. Furthermore, the business metadata will be developed for the output tables, which are the tables that result from the transformations. However, the scope of technical metadata includes metadata for all data sets in the SDZ and the GDZ.



SCOPE OF BUSINESS METADATA DEVELOPMENT

The business metadata will at least be developed for the output tables, which are the tables that result from the transformations. The attributes in the metadata template must be equal for both business metadata and technical metadata after distinction to accommodate the accessibility of the user.



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KEY FIELDS IN BUSINESS METADATA TEMPLATE

No.	Business Attribute Name	Business Definition	Column Name	PII/ Non-PII	List of Allowable Values	Sample Data	Business Data Steward	Data Owner	Updated Date	Updated By	Additional Remarks	Subdomain
24	Customer Phone Number	Customer Phone Number represents the phone number of the customer.	PHN_NUM	PII		0661011212 0412934563	Rewat Trangansri	Pasd Putthapipat	2023-01-31	Prin.I	Business Attribute Name, Business Definition, PII and Column Name updated	Customer Profile

No	Fields	Explanation
1	No	Running number
2	Business Attribute Name	Business-defined name of data attribute
3	Business Definition	Business meaning of this data attribute. The business attribute name and business definition must have a 1:1 mapping of each subsidiary
4	Column Name	Name of the data column in the data set. The column name will be the unique ID for business metadata of each subsidiary. The same business attribute name cannot occur in multiple rows within each monoline. The column name will be used for linkage with technical metadata.
5	PII/ Non-PII	Provides the information if the data attribute name is PII or Non-PII, supports PDPA requirement
6	List of Allowable Values	Finite list of allowable values that can be entered for the column name
7	Sample Data	Few example values that can be found and represent the characteristics of the data attribute
8	Business Data Steward	Contact of the Business Data Steward who is responsible to validate and verify business term, business definition and data set. This can include multiple Business Data Steward from GDZ/SDZ
9	Data Owner	Contact of Data Owner who is responsible for the functionality and use of the data at the data subject area level. This can include multiple Data Owners from GDZ/SDZ
10	Updated Date	The date when latest change is registered to business metadata, format: dd/mm/yyyy
11	Updated By	Person who registered the latest changes on the business metadata
12	Additional Remarks	Description of the changes made in the business metadata
13	Subdomain	Subdomain the grouping of data and area of interest within subsidiary

KEY FIELDS IN TECHNICAL METADATA TEMPLATE

System/ Data Subject Area Name	Schema/Database Name	Data Set Name	Column Name	Data Zone (MDZ/GDZ)	PII/ Non-PII	Type	Format	Length	Nullable	Updated Date	Updated By	Additional Remarks
Unify Customer Mapping	{menv}_MDM_UCID_DB	UCID_MPNG_CUR	UC_ID	GDZ	PII	BIGINT	Numeric	19 digits	No	2023-01-31	Prin.I	

No	Fields	Explanation
1	System/ Data Subject Area Name	Name of system in persist zone or data subject area name in curate zone
2	Schema/ Database Name	Schema/ Database can be defined as a collection of tables and views of tables – this information includes environment, persist/ curate zone , system name/data subject area
3	Data Set Name	Data set is a collection of information within the schema/ database e.g., table name
4	Column Name	Name of the data column in the data set.
5	Data Zone (MDZ/ GDZ)	Name of the data zone from where the source system information is captured : SDZ, GDZ
6	PII/Non-PII	Provides the information if the data attribute name is PII or Non-PII, supports PDPA requirement
7	Type	Data type of the column name / data attribute name
8	Format	Format of the data field. For example – data type will have dd-mm-yyyy as the format
9	Length	Byte length of the data
10	Nullable	Can this field remain as blank: No, Yes
11	Updated Date	The date when latest changes are made in the system, format: dd/mm/yyyy
12	Updated By	Person who registered the latest changes in the system
13	Additional Remarks	Description of the changes made in the system (e.g., length is updated from 4 to 5)

KEY FIELDS IN DATA SET INVENTORY TEMPLATE

Data Set Name	Data Set Description	Data Set Update Frequency	Updated Date	Updated By	Additional Remarks
UCID_MPNG_DLY	Stored UCID number of every customer along with Deduplication Matched Score, which represents the matching confidence level, and Latest update UCID Rank Number, which is the rank of latest UCID for the customers from the same monoline that have the same UCID. This table collects the data in full dump type and data update frequency is daily. Estimated data available is T-1. The table unique by CUST_NUM for each record.	Daily	2023-02-09	Prin.I	Data Set description updated

No	Fields	Explanation
1	Data Set Name	Data set is a collection of information within the schema/ database e.g., table name
2	Data Set Description	Description of the table name in the source system
3	Data Set Update Frequency	The set frequency to update data set
4	Updated Date	The date when latest changes are made in the data set inventory template
5	Updated By	Person who registered the latest changes in the data set inventory template
6	Additional Remarks	Description of the changes made in the data set inventory template (e.g., data set update frequency changed from monthly to weekly)

KEY GUIDELINES TO DEFINE BUSINESS METADATA

2 Business Attribute Name

No	Fields	Explanation
2	Business Attribute Name	Business-defined name of data attribute

Modifier Word

is an adjective that further refines a business attribute name or adjective used to describe the main noun.

- Short
- Long
- Current
- Year-to-date
- Primary
- Secondary
- Alternate

Prime Word

is the main noun that is the subject of a business attribute name.

- **Party Roles:** Employee, Contractor, Customer
- **Places:** District, Branch, State
- **Things:** Equipment, Product, Document
- **Organizations:** Department, Vendor, Business Unit
- **Concepts and Ideas:** Task, Account, Objective
- **Events:** Transaction, Phase Receipt

Class Word

is a word that classifies a business attribute name by its function.

* Business Attribute Name should end with class word

- Address
- Amount
- Balance
- Code
- Date
- Description
- Identifier
- Flag
- Name
- Number
- Quantity
- Count
- Rate
- Timestamp
- Percent
- Ratio
- Status
- List
- Score

KEY GUIDELINES TO DEFINE BUSINESS METADATA

3 Business Definition

No	Fields	Explanation
3	Business Definition	Business meaning of this data attribute. The business attribute name and business definition must have a 1:1 mapping of each subsidiary

Business Definition

The format should be "Business attribute name **refers to / represents / is** XX" where XX is the business context provided in easy-to-understand language without any abbreviations.

Length

If there exists a known fixed length of data, then it should be called out as "Its length should be XX"

Format

If there exists a format of values for the business attribute name, then it should be called out as "The format should be XX" where XX is the format.

For example, date format is 'YYYY-MM-DD'

Current/Sample list of allowable values

- If list of allowable value is infinite: not be included
- If less than 5 allowable values: include all values in business definition
- If equal to or more than 5 allowable values: include 5 values by using the sentence "Please refer to link of allowable values"

Note(s)

If additional clarification needs to be provided, it should be written as "Note(s) - XX" where XX is the additional information.

KEY GUIDELINES TO DEFINE BUSINESS METADATA

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List of Allowable Values

No	Fields	Explanation
6	List of Allowable Values	Finite list of allowable values that can be entered for the column name

Type	List of Allowable Values (Link)
Flag Type contains only "Yes" and "No" allowable value	Link of allowable value is not necessary
Others contains allowable values besides "Yes" and "No"	List of allowable value to be created to provide a navigation link to a separate worksheet where all the allowable values and their details are provided

If there exists a finite list of allowable values, then please refer the following table:

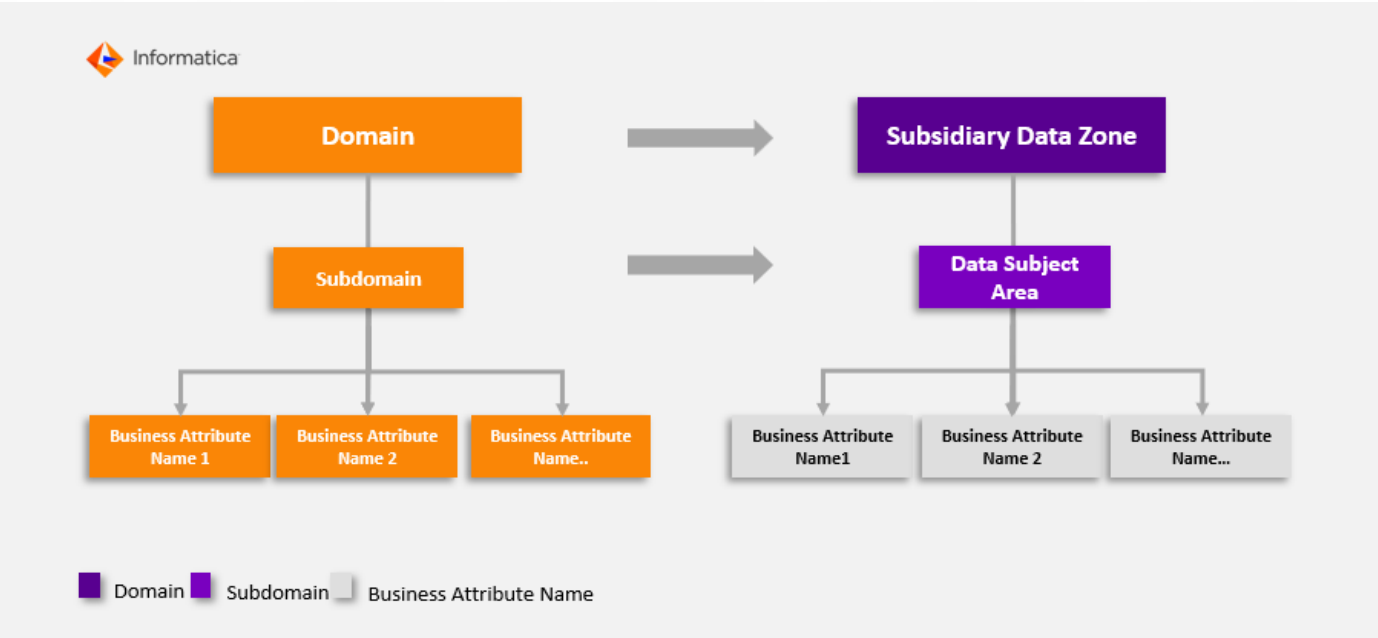
Scenario for "Current List of Allowable Values"	Guidance for business definition
Less than 5 allowable values	Include all values in business definition Format: "Current list of allowable values include: Value 1 Value 2 Value 3 etc."
Equal to or more than 5 allowable values	Include 5 values followed by the link to complete list by using the sentence "Please refer to link of allowable values for complete list of allowable values" Format: "Sample list of allowable values include: Value 1 Value 2 Value 3 Value 4 Value 5. Please refer to link of allowable values for complete list of allowable values."

KEY GUIDELINES TO DEFINE BUSINESS METADATA

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Subdomain

No	Fields	Explanation
13	Subdomain	Subdomain the grouping of data and area of interest within subsidiary



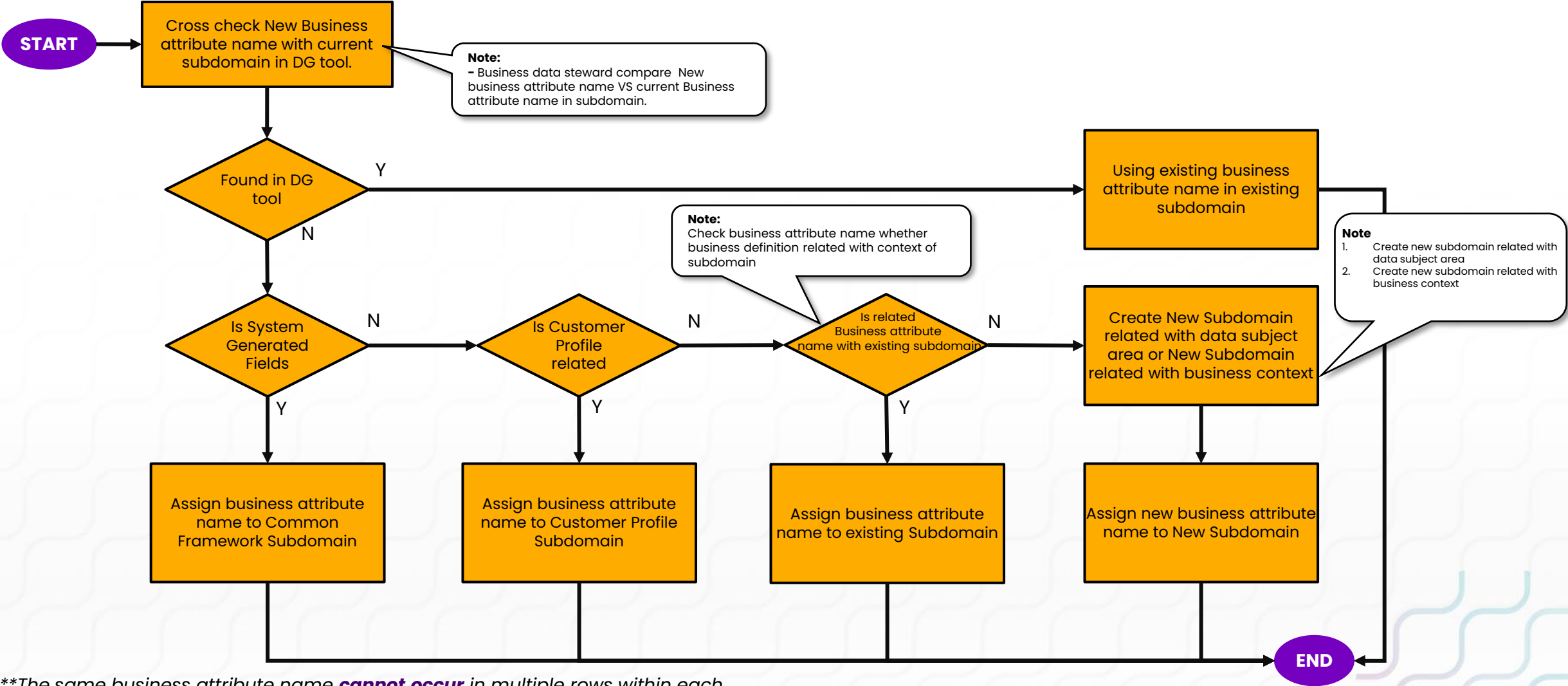
Domain

Domain is the highest level of grouping which we use for representing **group of data from subsidiaries** of **SCBX**.
e.g., for DataX will be setup domain as **GDZ**, for AutoX monoline will be setup domain as **AutoX**.

Subdomain

Subdomain provide a logical grouping of data and area of interest within organization that we define based on **data subject area**.
However, business data steward able to consider to create/manage **another subdomain** that can help to organize the data in order to fit with the business context.

FLOW TO CREATING SUBDOMAIN AND ASSIGNING BUSINESS ATTRIBUTE NAME



***The same business attribute name **cannot occur** in multiple rows within each subsidiary



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Overview of metadata management

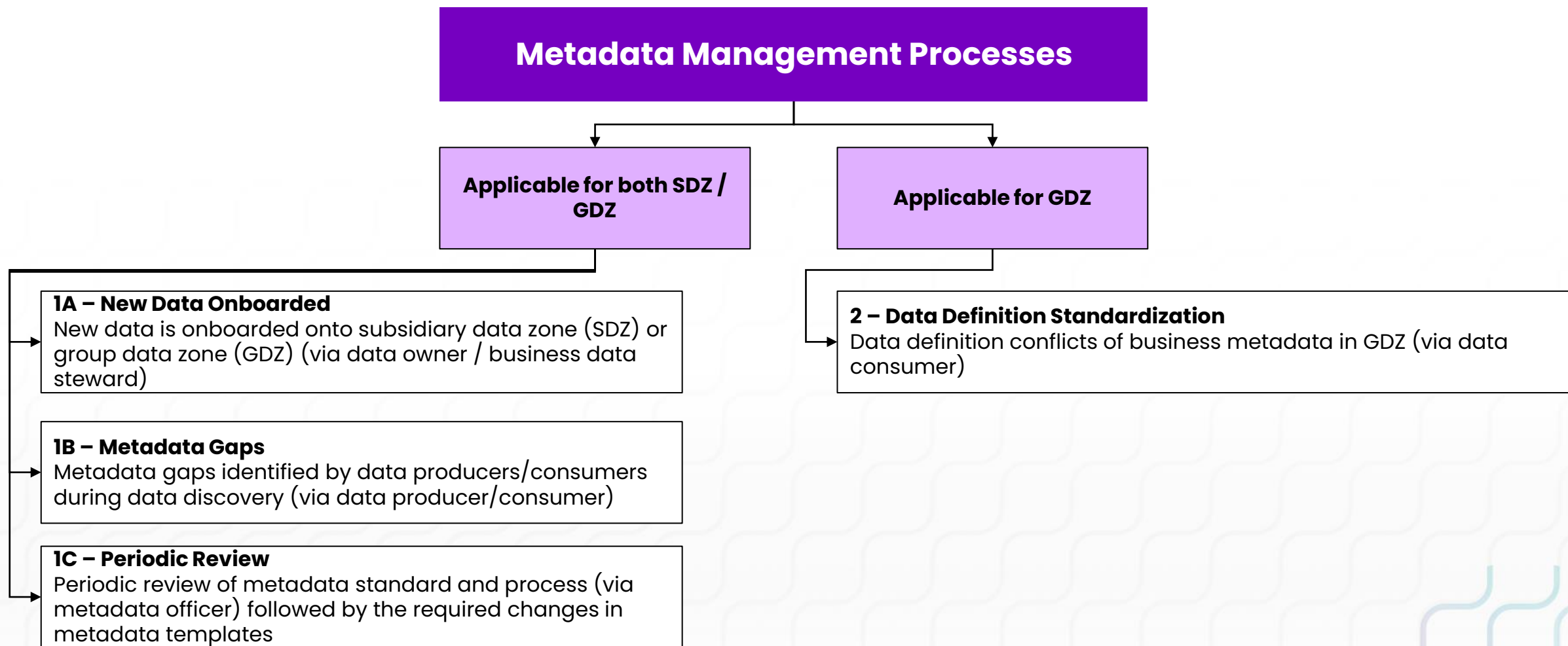
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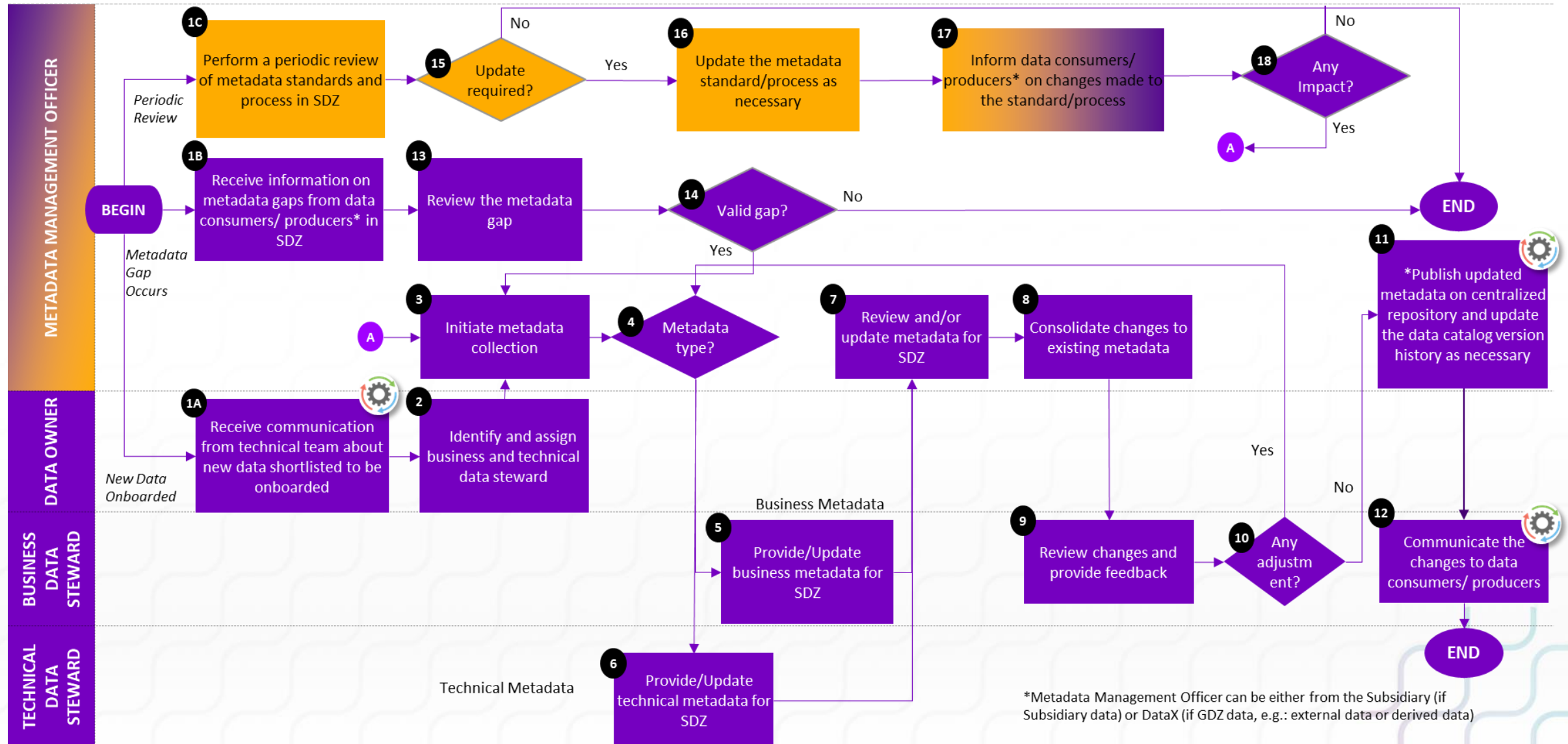
Metadata management processes

METADATA MANAGEMENT PROCESS – INTRODUCTION





CREATE, UPDATE AND MAINTAIN METADATA PROCESS



Thank You

Please reach out to
datax_metadata@data-x.ai for more
information.